



STM Exercise 5

Phase Diagrams

(full solubility in the solid state)

(no solubility in the solid state)

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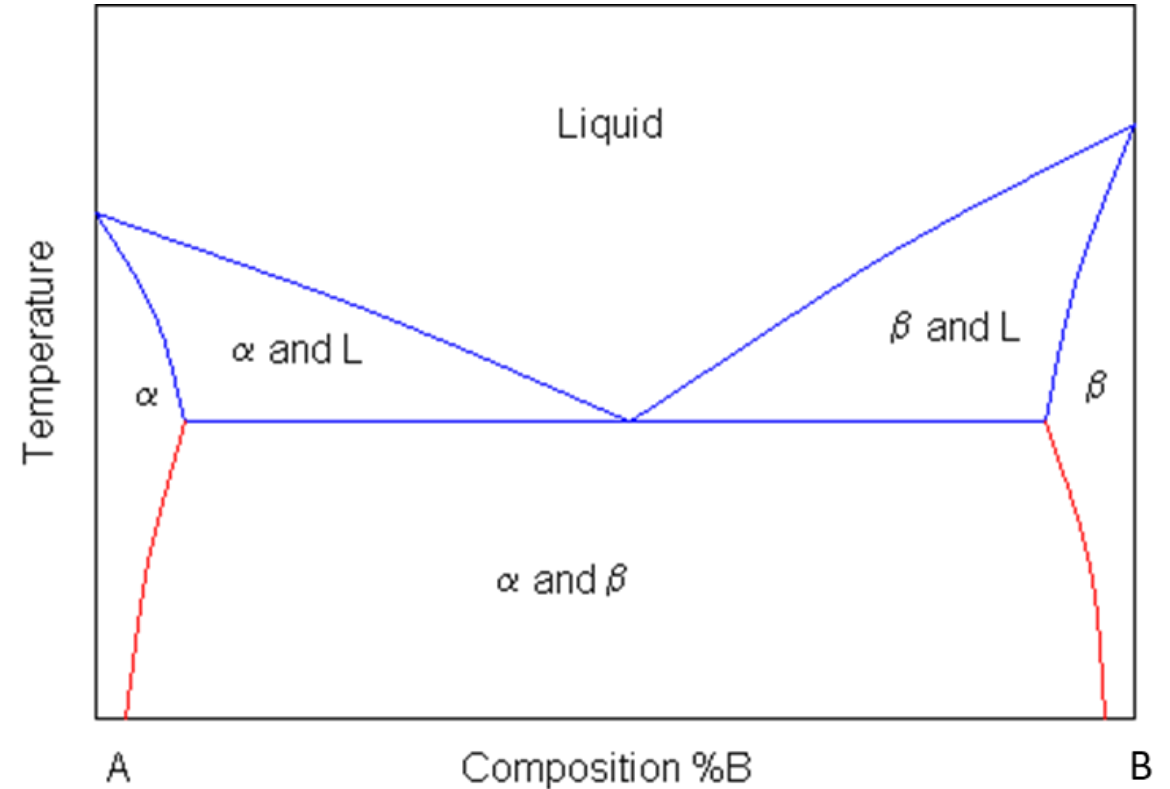
Phase Diagram

it shows the equilibrium states of a mixture, so that given a temperature and composition, it is possible to calculate which phases will be formed, and in what quantities.

Where **L** stands for liquid, and **A** and **B** are the two components and **α** and **β** are two solid phases rich in A and B respectively.

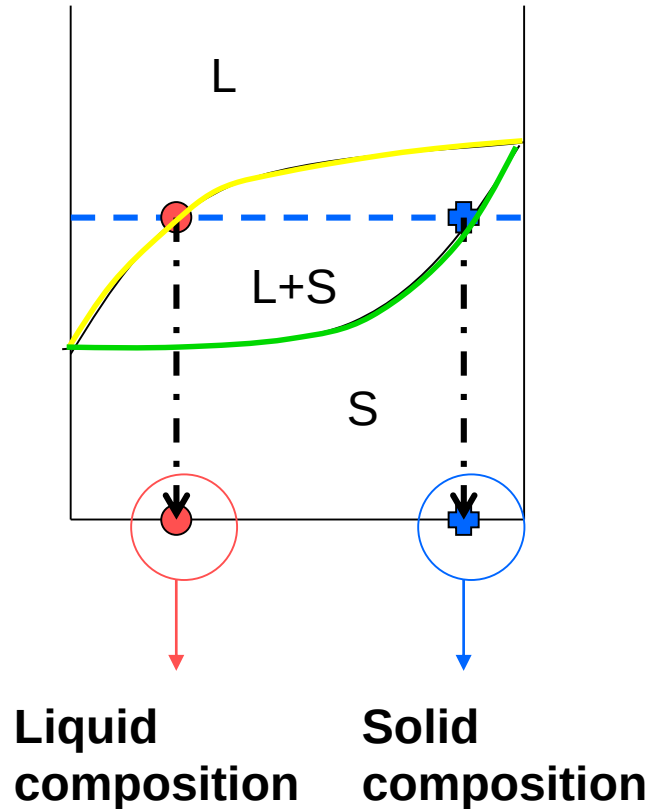
The blue lines represent the liquidus and solidus lines

The red lines involve a solid-to-solid transition (solvus lines)



Determination of the compositions of phases in equilibrium in a biphasic zone:

HORIZONTAL RULE



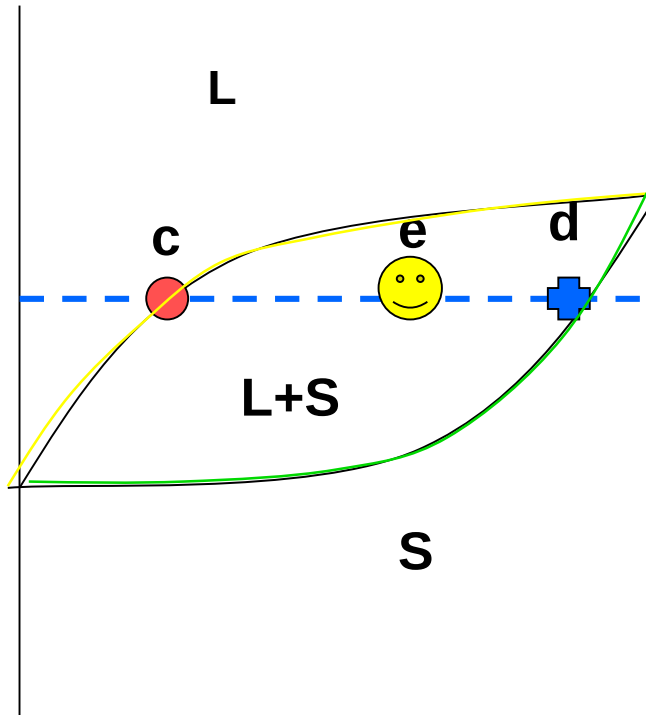
- Draw the tie line - - - - -
- Individuate intersections with **liquidus** (•) and **solidus** (+) lines
- The **composition of the liquid** is the value on the x axis correspondent to the intersection between the tie line and the liquidus line.
- The **composition of the solid** is the value on the x axis correspondent to the intersection of the tie line with the solidus line.

LIQUIDUS CURVE = above it only liquid is present

SOLIDUS CURVE = below it only solid is present

Phases relative amounts determination in a biphasic zone, at equilibrium:

LEVER RULE



- Draw the tie line - - - - -
- Individuate the intersection between tie line and liquidus line (• c) and solidus line (⊕ d)
- The % of phases in a point of the biphasic region (e) is defined as:

$$\%L = (ed/cd) \cdot 100$$

$$\%S = (ce/cd) \cdot 100$$

The lever rule

If an alloy consists of more than one phase, the amount of each phase present can be found by applying **the lever rule** to the phase diagram.

The lever rule can be explained by considering a simple balance.

The composition of the alloy is represented by the fulcrum C , and the compositions of the two phases by the ends of a bar (C_1 and C_2).

Each phase are determined by the weights needed to balance the system.

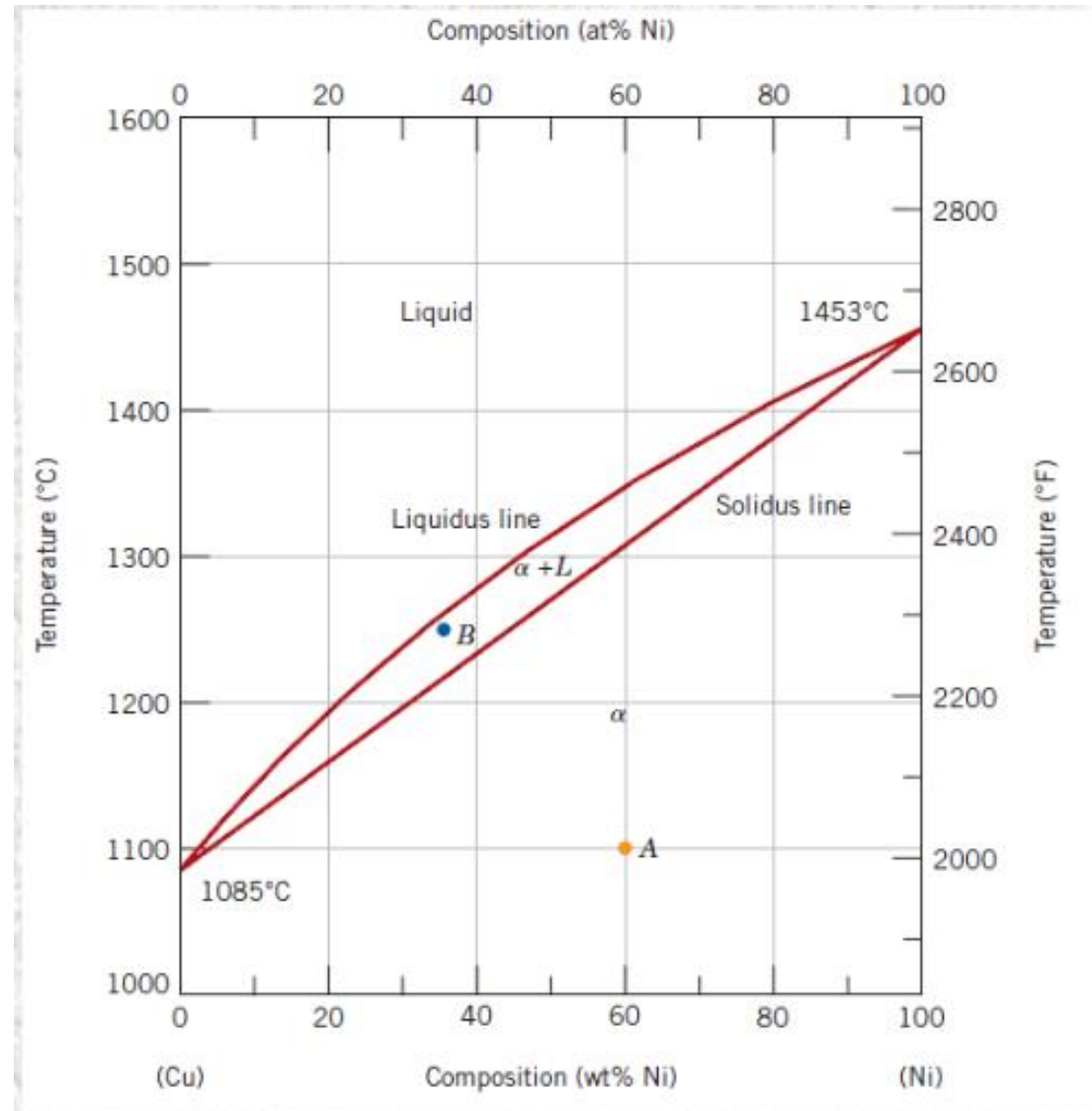
$$\text{Fraction of phase 1} = (C_2 - C) / (C_2 - C_1)$$

$$\text{Fraction of phase 2} = (C - C_1) / (C_2 - C_1).$$

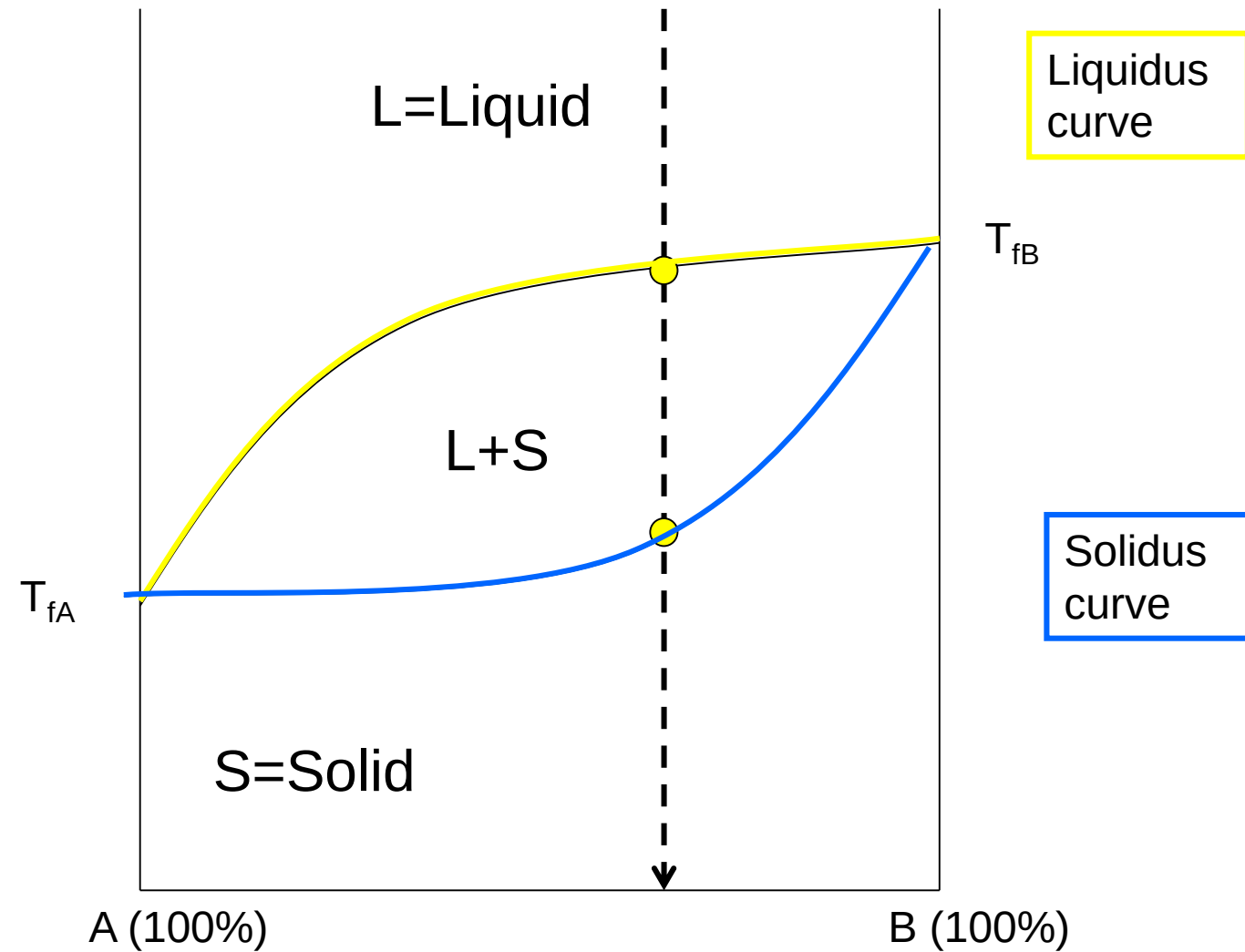


Exercise 1: Draw a phase diagram (for generic components A and B), with complete solubility in the liquid state and **complete solubility in the solid state**. Also indicate the phase/phases present(s) in each region of the phase diagram

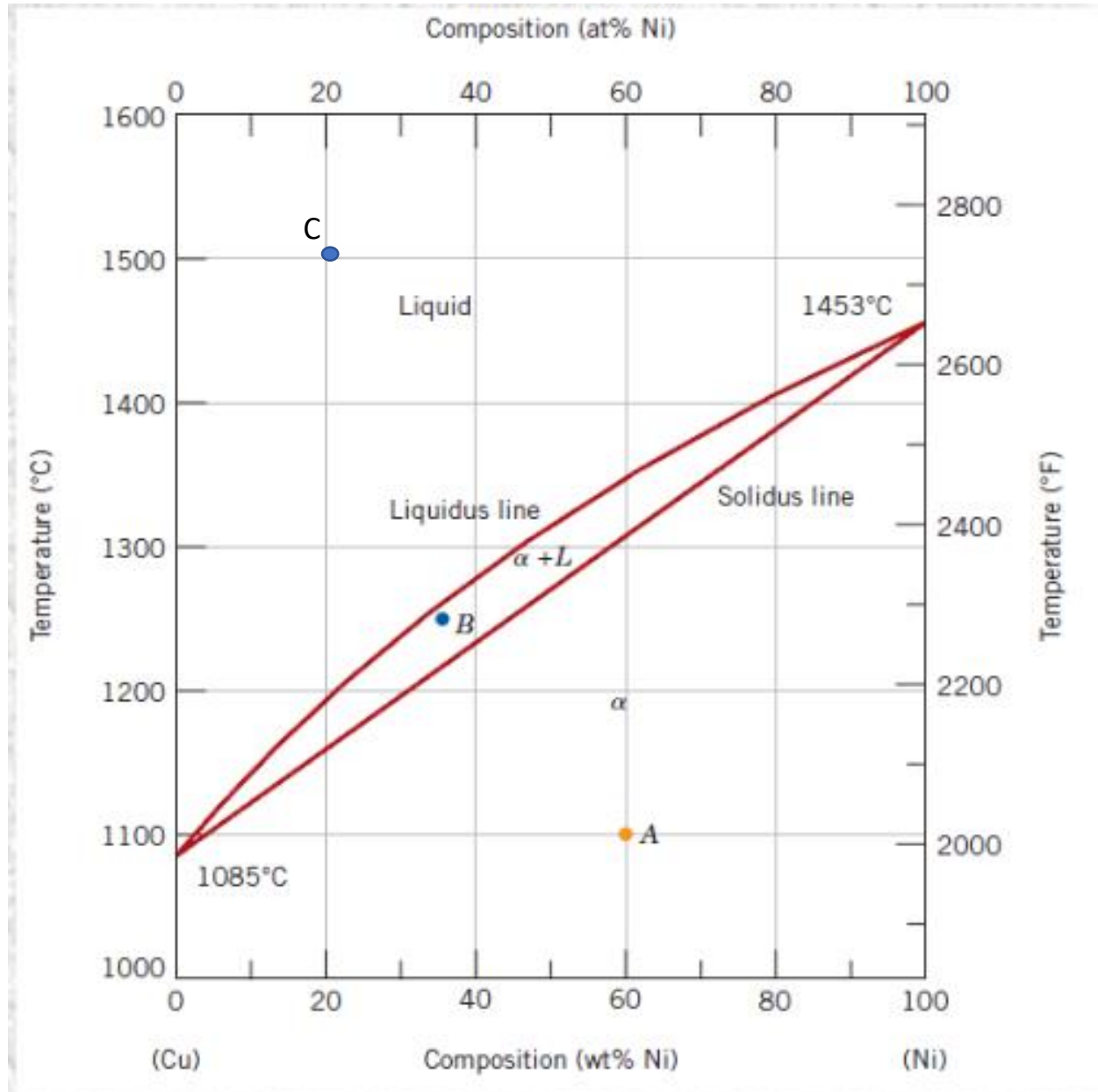
Exercise 1: full solubility in the solid state

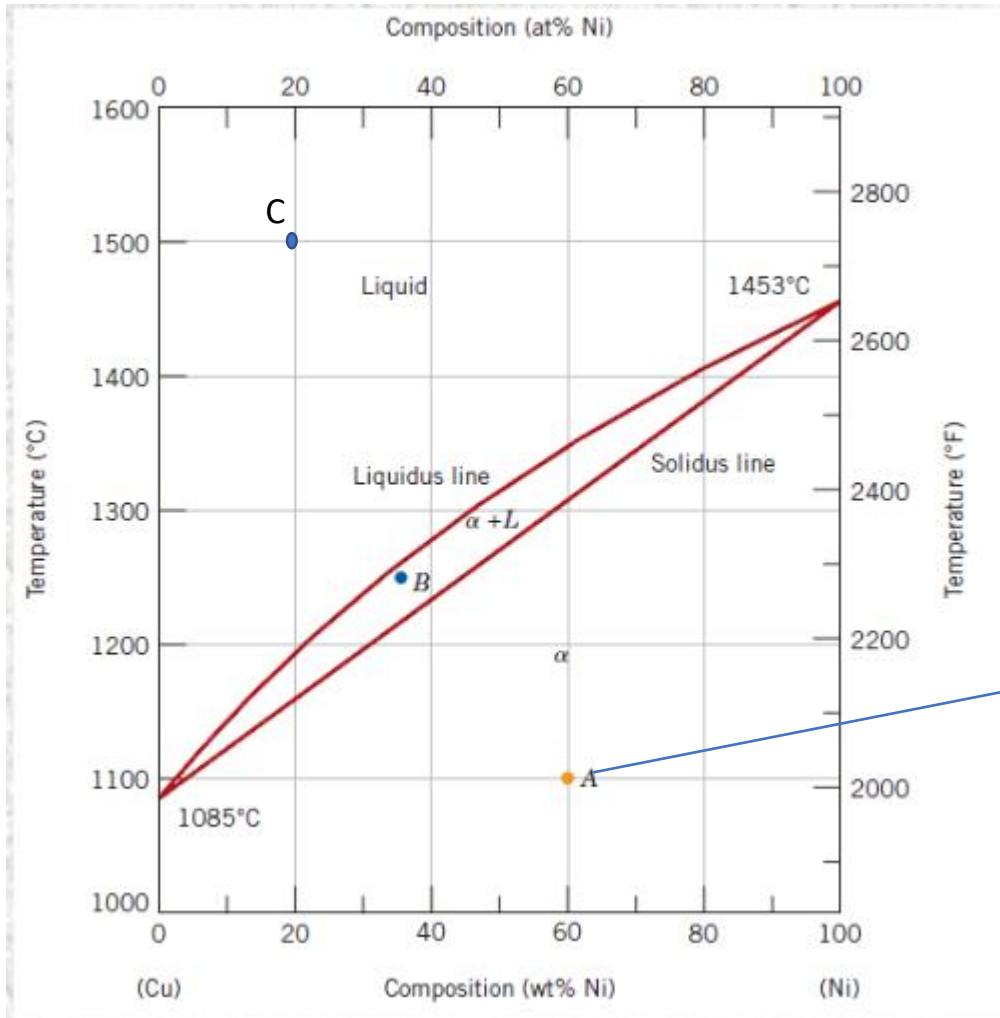


FULL SOLUBILITY IN THE SOLID AND LIQUID STATE



Exercise 2: determine the phases present, their composition and relative amounts in the point A, B and C indicated in the phase diagram. Schematize the microstructure in each point.



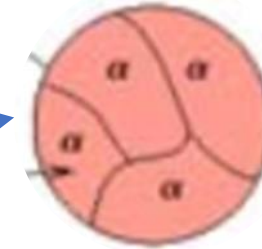


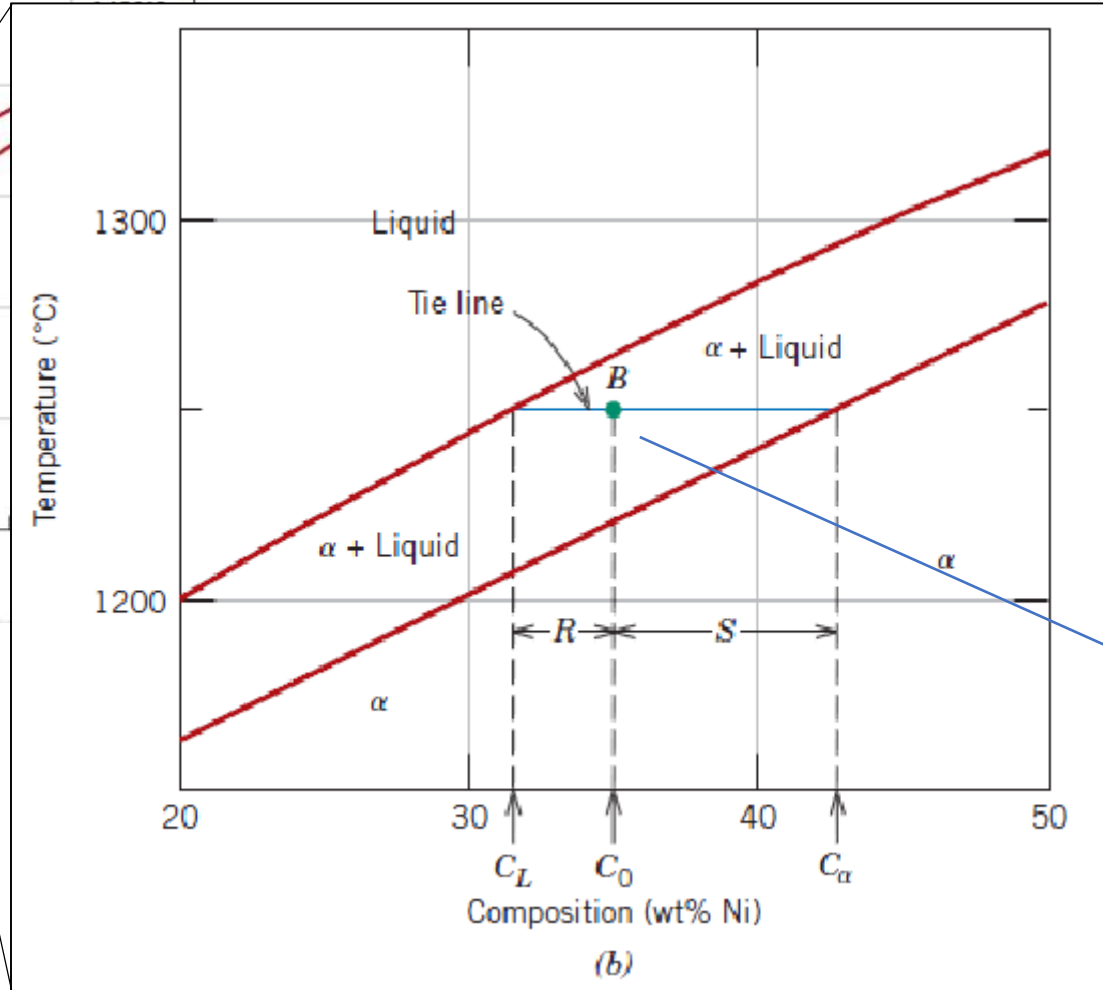
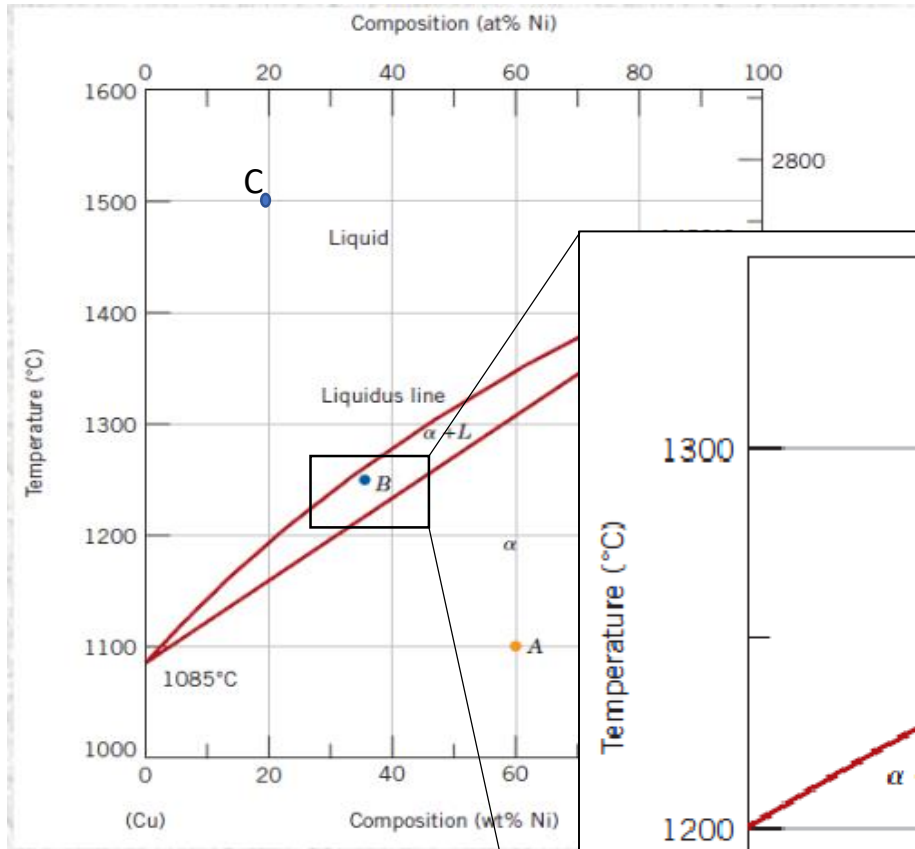
Point A:

Phases: 1 phase solid

Composition: 60%Ni, 40% Cu

Relative amount: 100% solid phase





Point B:

Phases: 2 phases solid (α) + liquid (L)

Composition: 2 phases region → horizontal rule

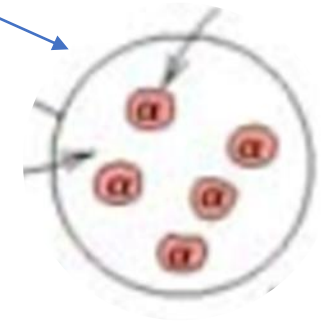
Composition of liquid (L): $C_L = 32\% \text{ Ni}, 68\% \text{ Cu}$

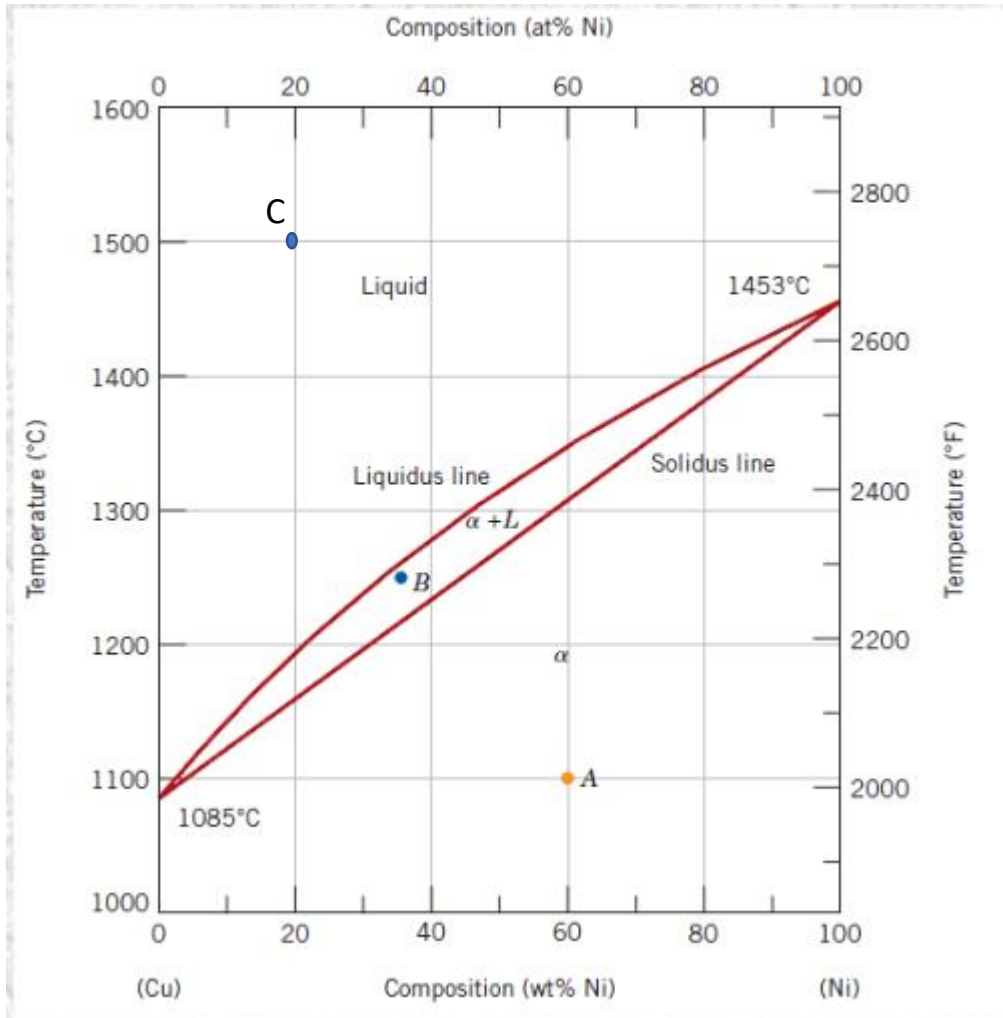
Composition of solid (α): $C_\alpha = 43\% \text{ Ni}, 57\% \text{ Cu}$

Relative amount: 2 phases region → lever rule

$\% \alpha = R / (R + S) * 100 = (35 - 32) / (43 - 32) * 100 = 27\%$

$\% L = S / (R + S) * 100 = (43 - 35) / (43 - 32) * 100 = 73\%$





Point C:

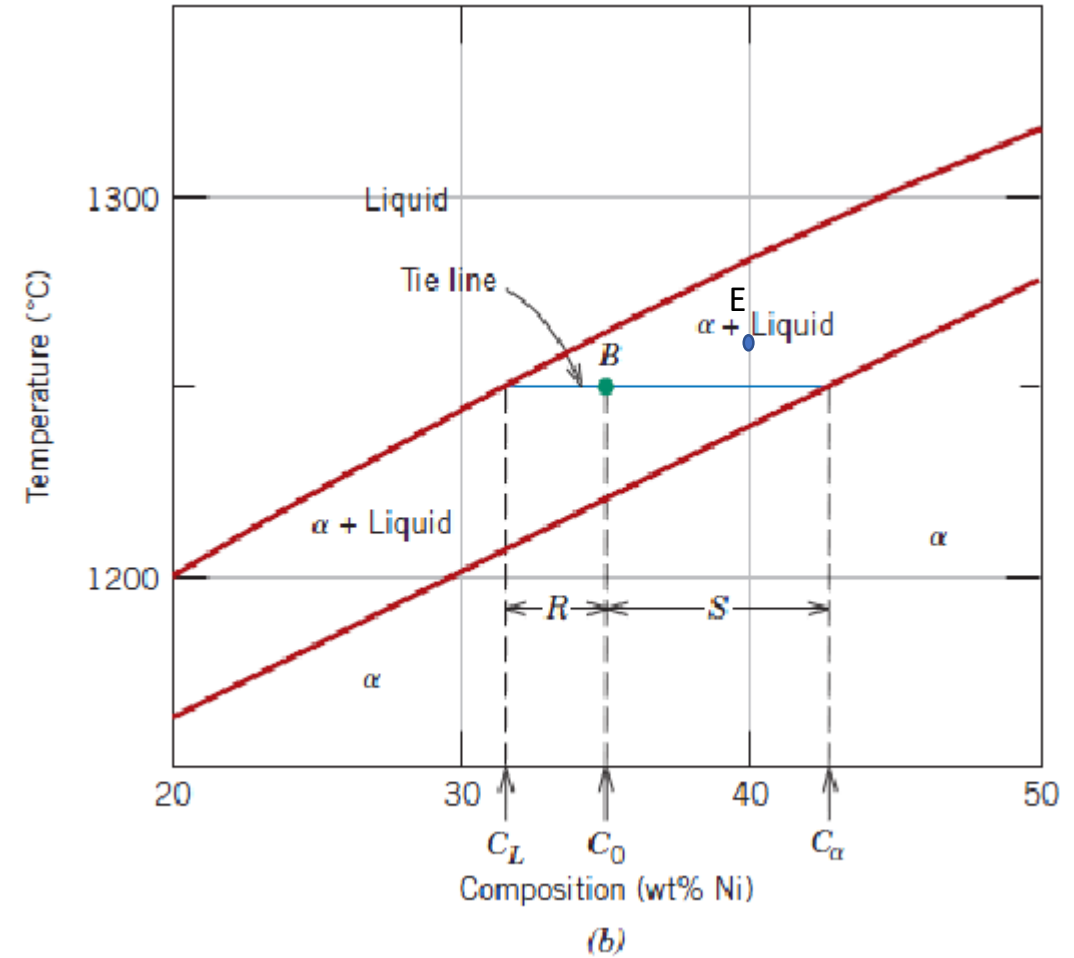
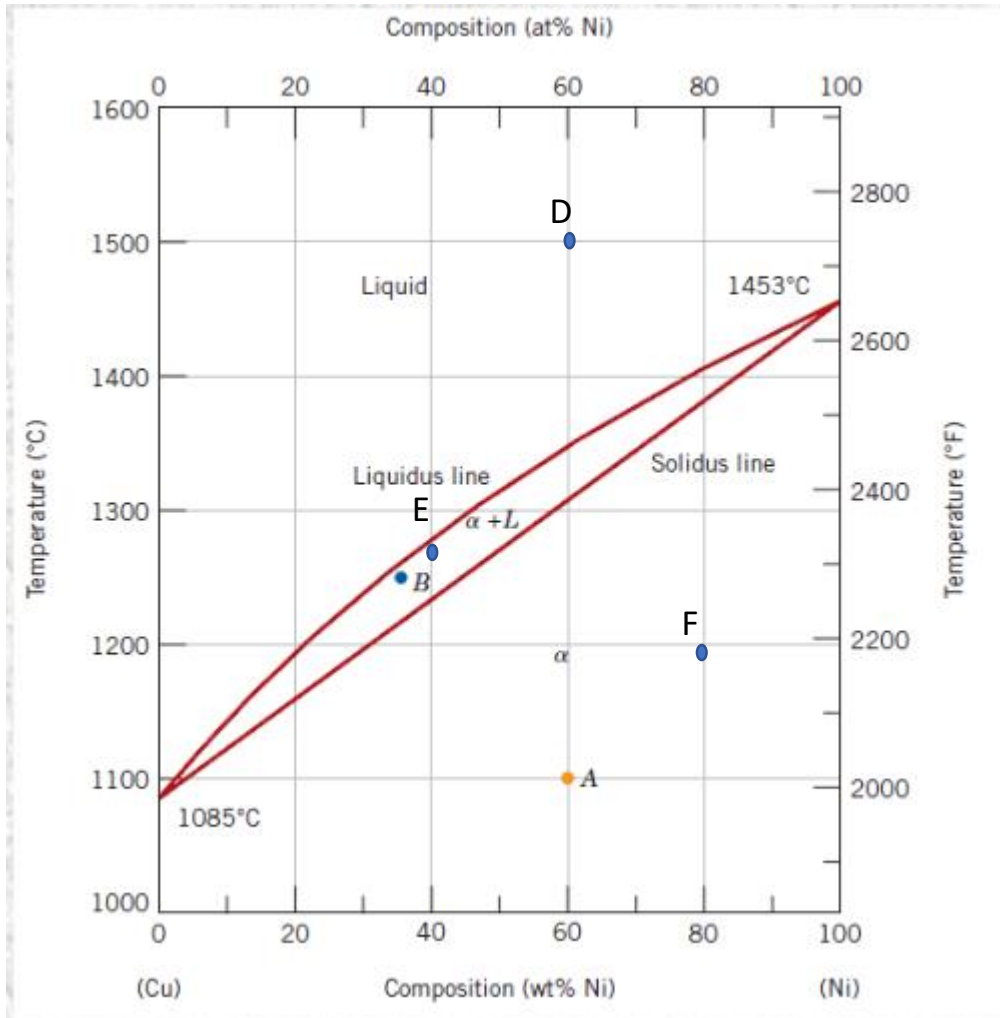
Phases: 1 phase liquid

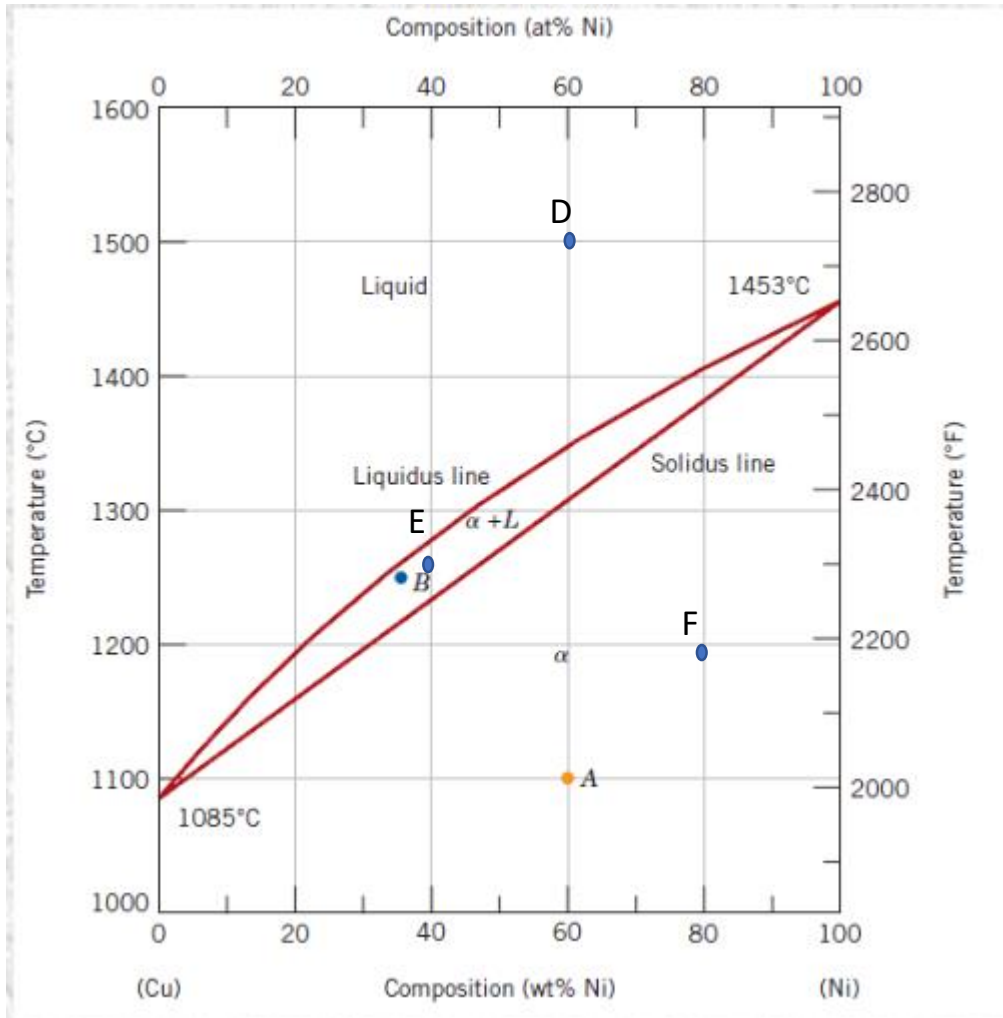
Composition: 20%Ni, 80% Cu

Relative amount: 100% liquid phase



Exercise 3: determine the phase present, their composition and relative amounts in the point D, E and F indicated in the phase diagram. Schematize the microstructure in each point.





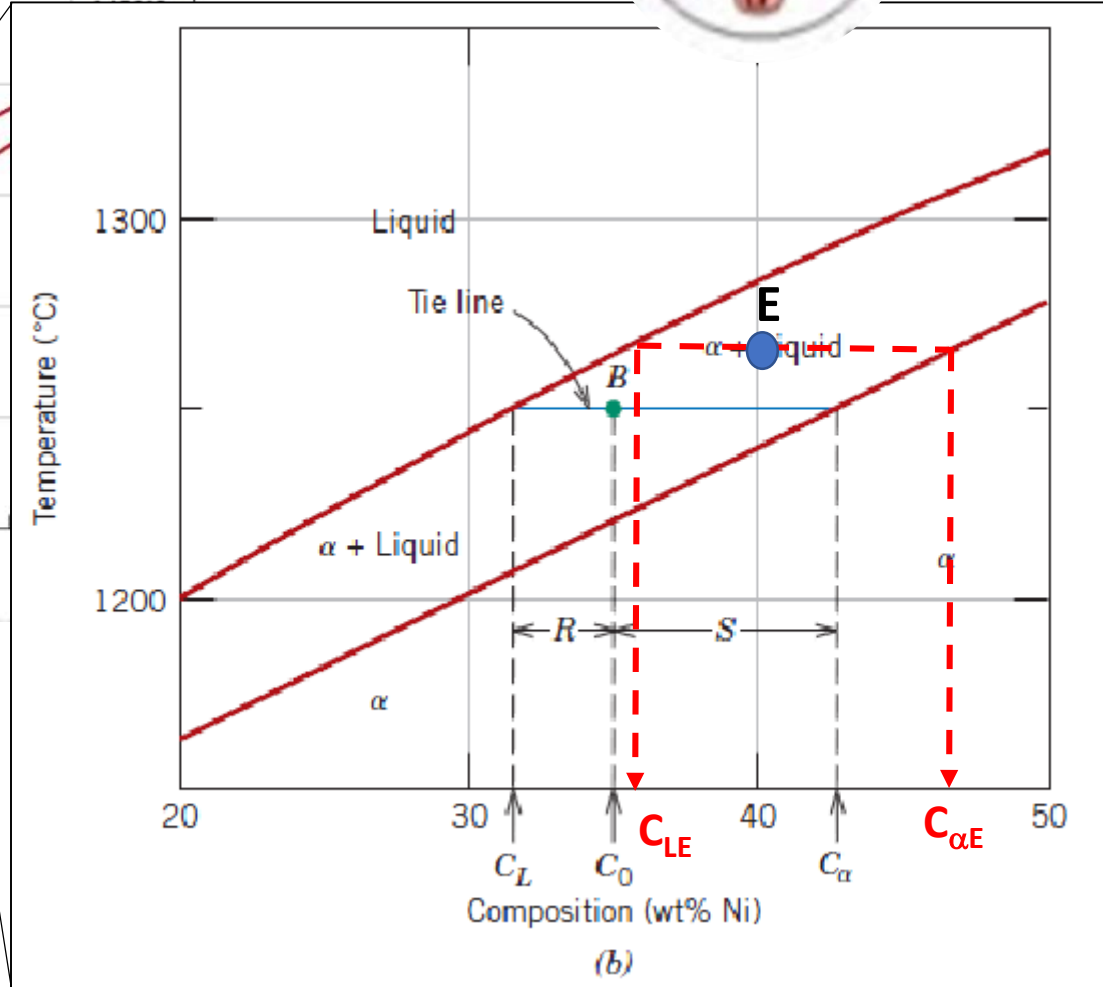
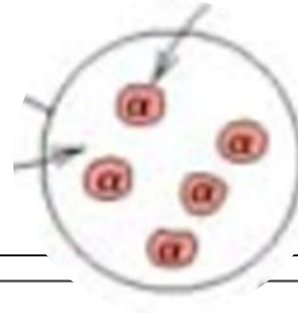
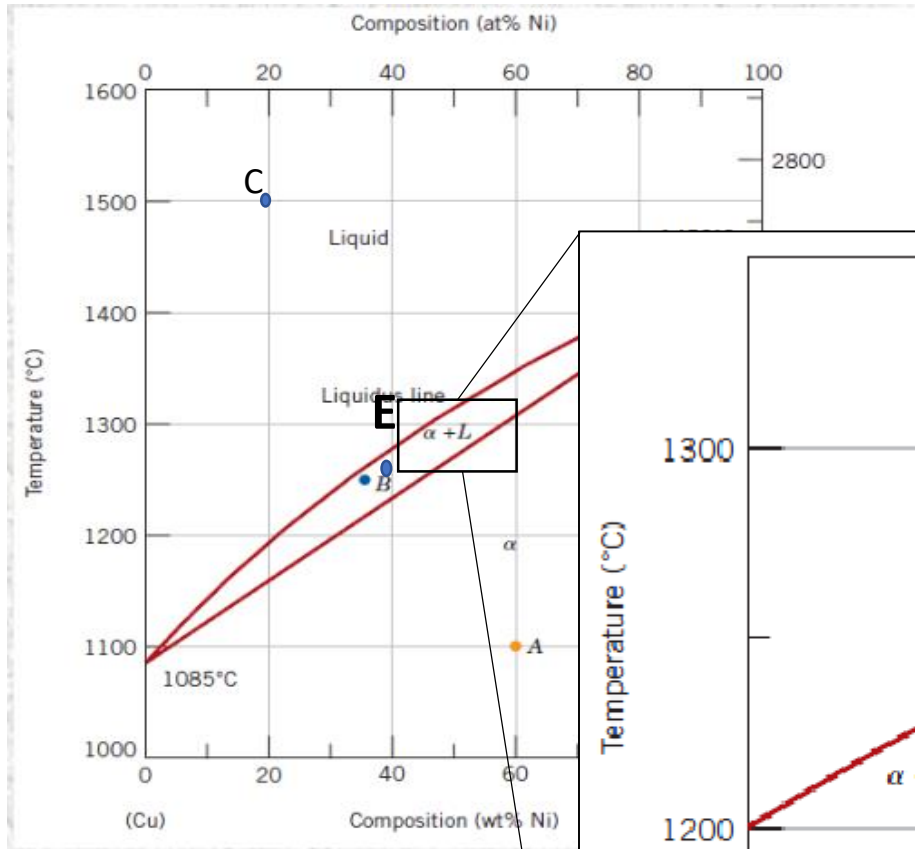
Point D:

Phases: 1 phase liquid

Composition: 60%Ni, 40% Cu

Relative amount: 100% liquid phase





Point E:

Phases: 2 phases = solid (α) + liquid (L)

Composition: 2 phases region → horizontal rule

Composition of liquid (L):

$C_{LE} = 36\% \text{ Ni}, 64\% \text{ Cu}$

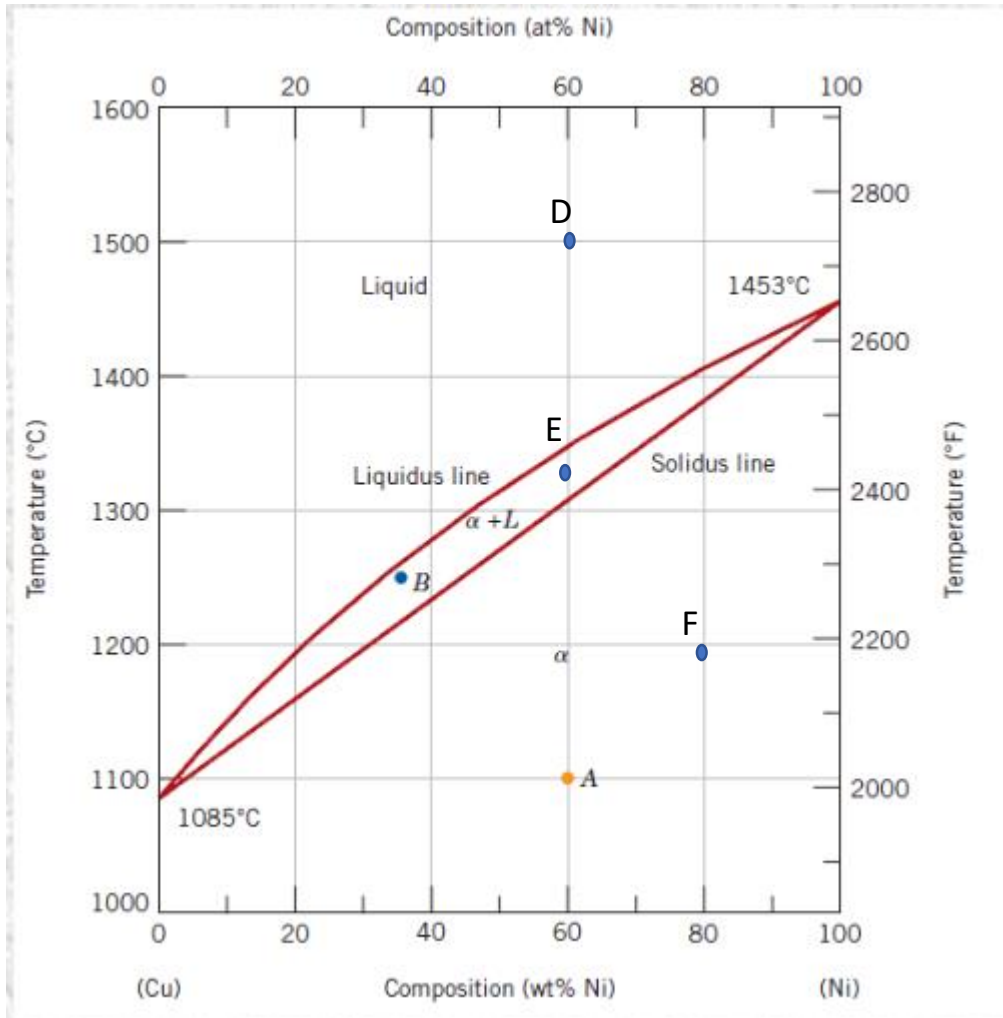
Composition of solid (α):

$C_{\alpha E} = 46\% \text{ Ni}, 54\% \text{ Cu}$

Relative amount : 2 phases region → lever rule

$\% \alpha = (40 - 36) / (46 - 36) * 100 = 40\%$

$\% L = (46 - 40) / (46 - 36) * 100 = 60\%$

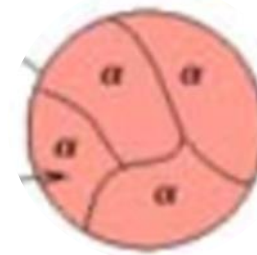


Point F:

Phases: 1 phase solid

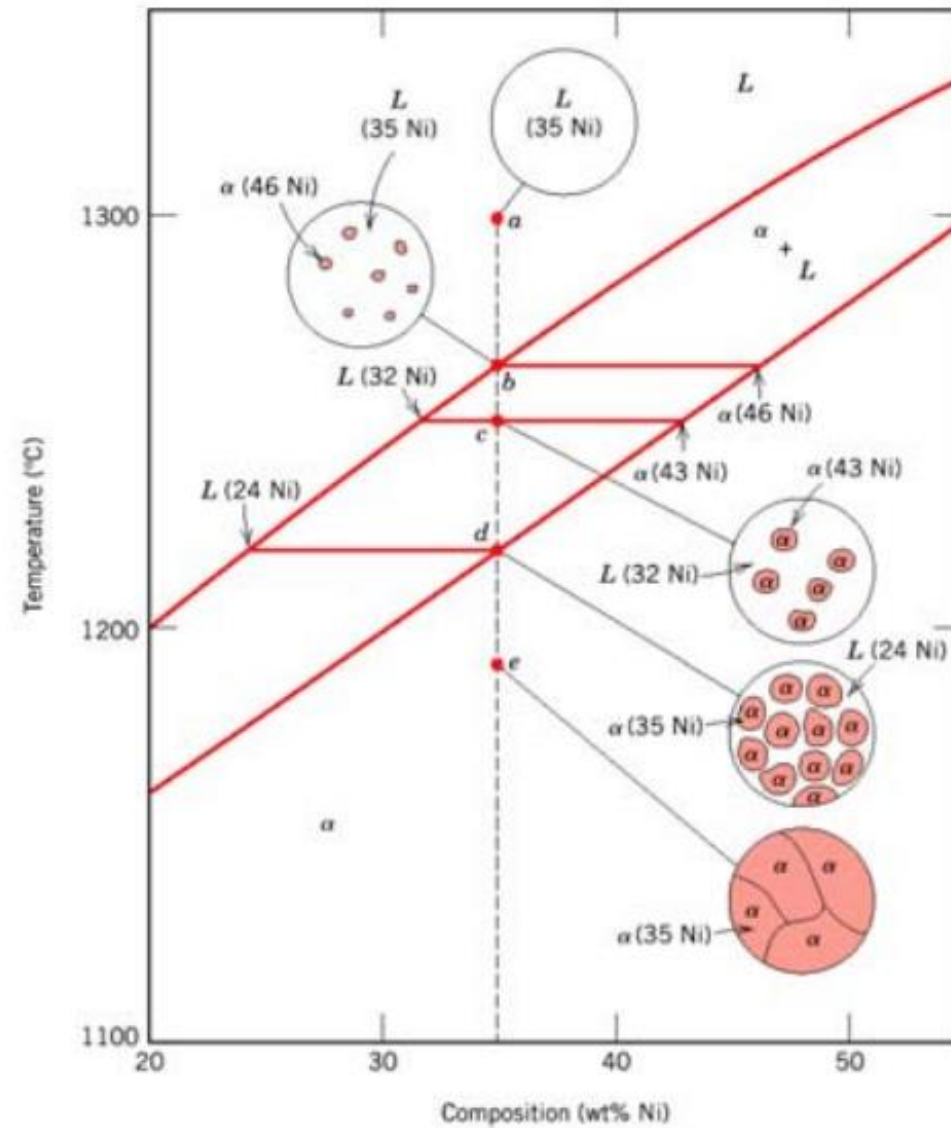
Composition: 80%Ni, 20% Cu

Relative amount: 100% solid phase



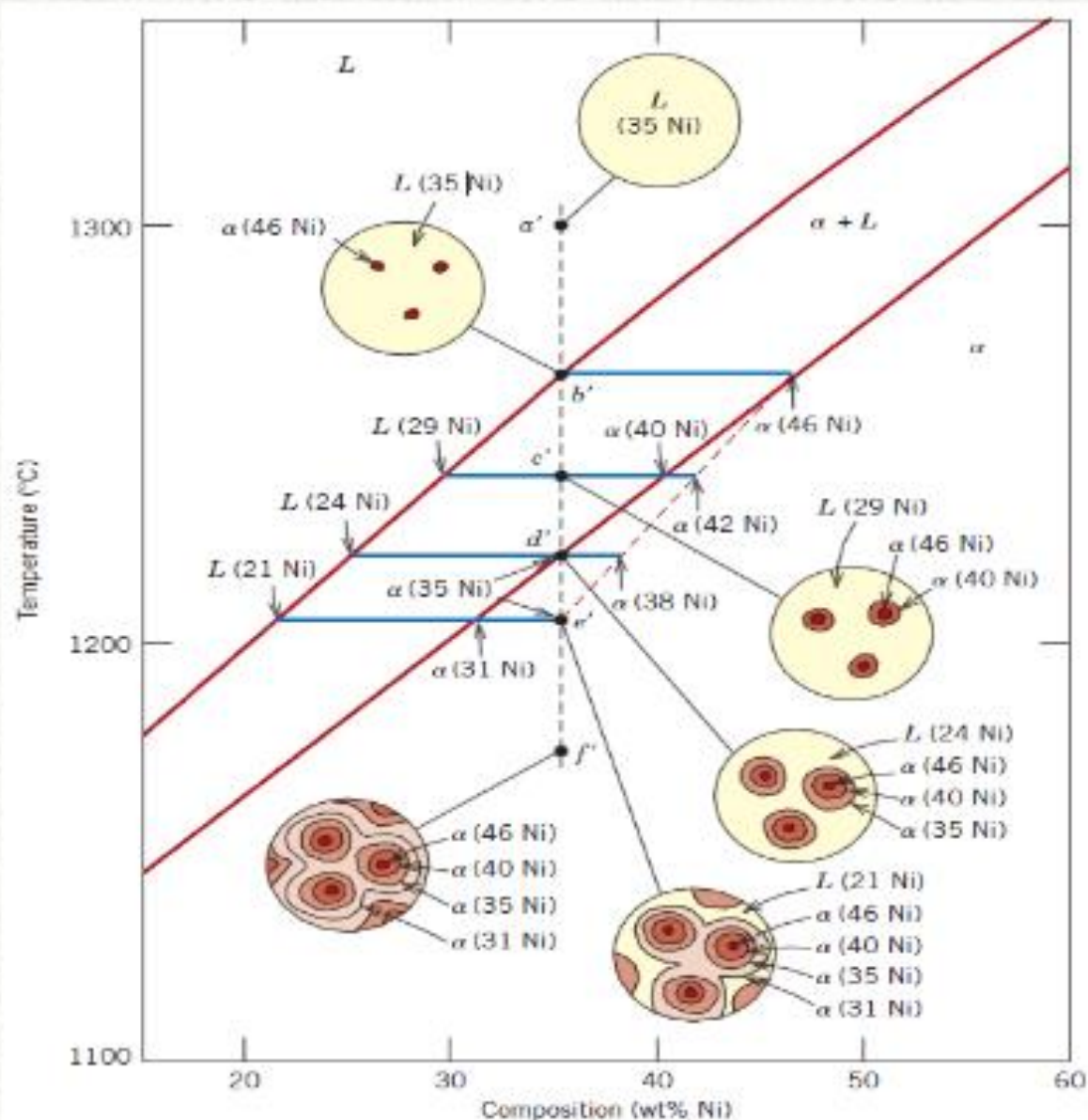
Development of microstructure in isomorphous alloys

Equilibrium (very slow) cooling

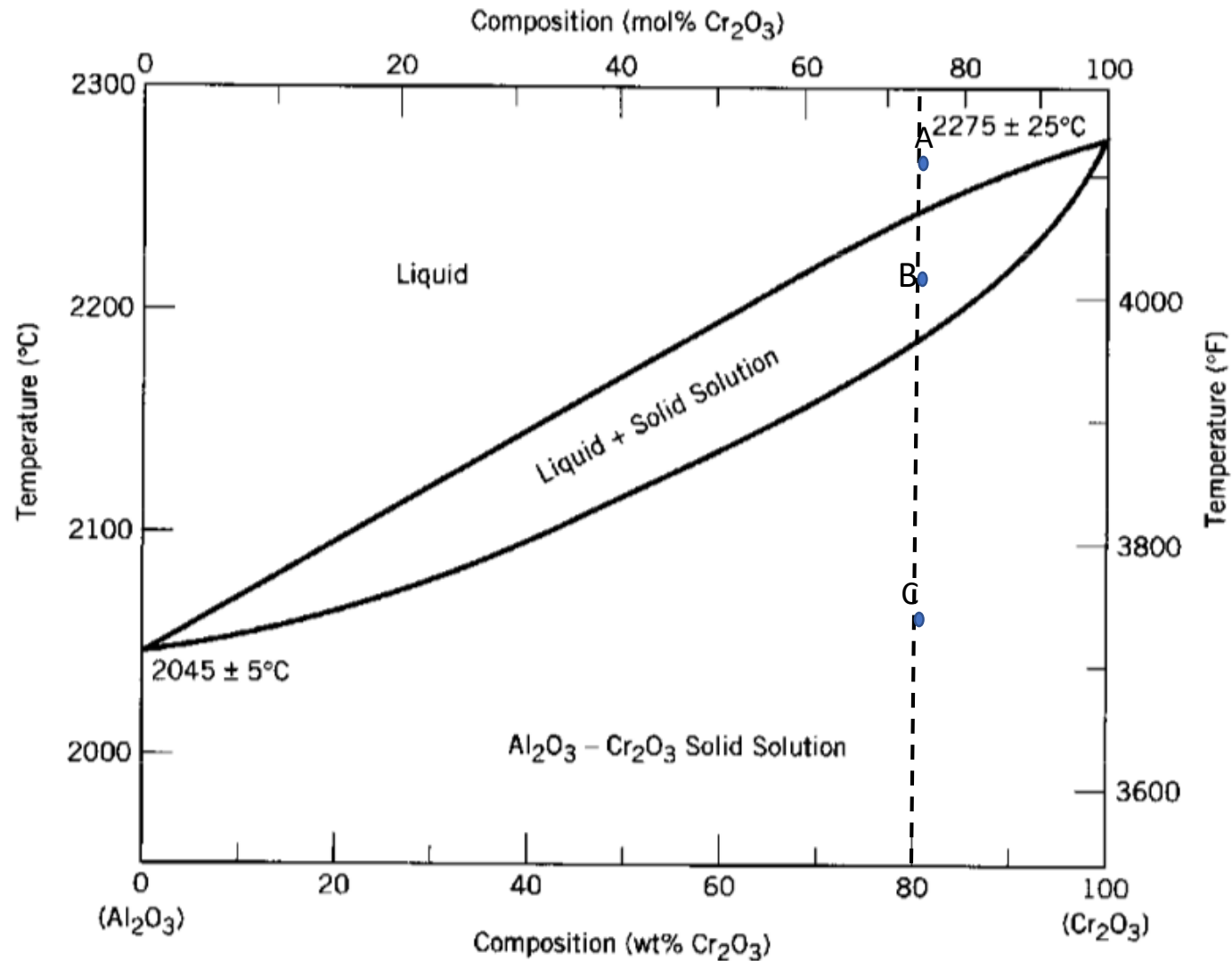


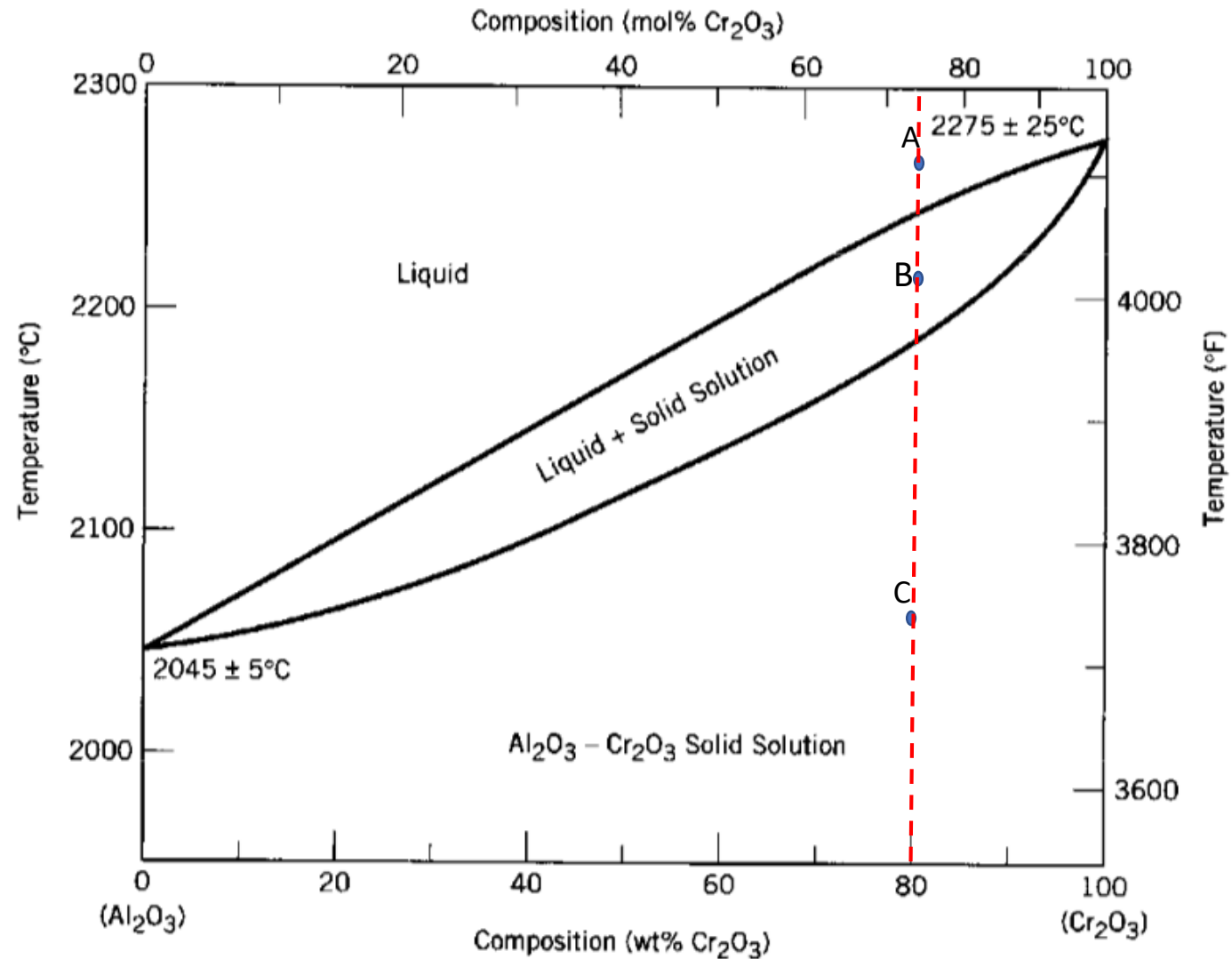
Non-Equilibrium Cooling

For Large cooling rate there is not enough time for diffusion!



Exercise 4: determine the phase present, their composition and relative amounts in the point A, B and C indicated in the phase diagram.



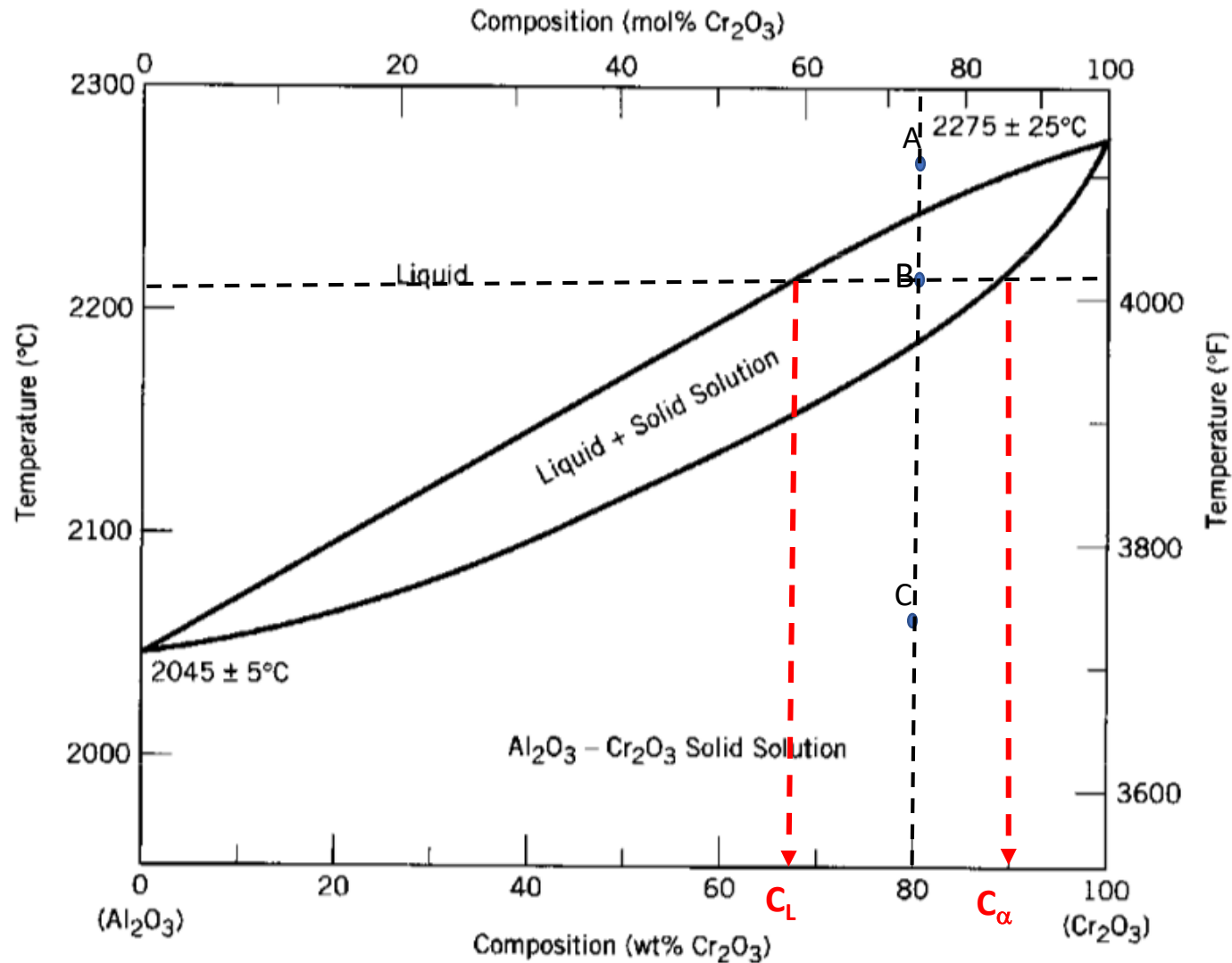


Point A:

Phases: 1 phase liquid

Composition: 80% Cr_2O_3 , 20% Al_2O_3

Relative amount: 100% liquid phase



Point B:

Phases: 2 phases solid (α) + liquid (L)

Composition: 2 phases region $\rightarrow$ horizontal rule

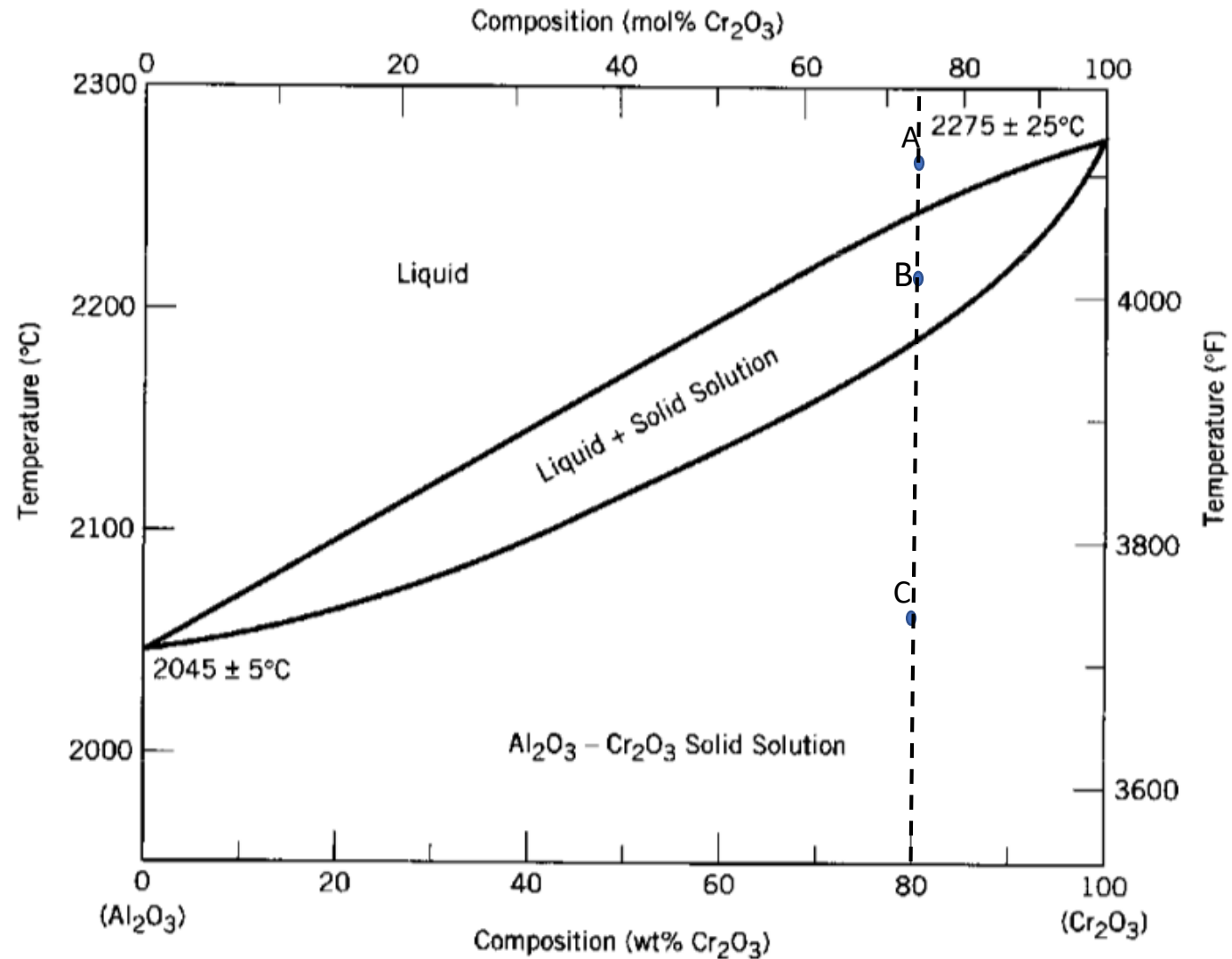
Composition of liquid (L):
 $C_L = 68\% \text{ Cr}_2\text{O}_3, 32\% \text{ Al}_2\text{O}_3$

Composition of solid (α):
 $C_{\alpha} = 90\% \text{ Cr}_2\text{O}_3, 10\% \text{ Al}_2\text{O}_3$

Relative amount: 2 phases region $\rightarrow$ lever rule

$$\% \alpha = \frac{(80 - 68)}{(90 - 68)} \times 100 = 54.5\%$$

$$\% L = \frac{(90 - 80)}{(90 - 68)} \times 100 = 45.5\%$$



Point C:

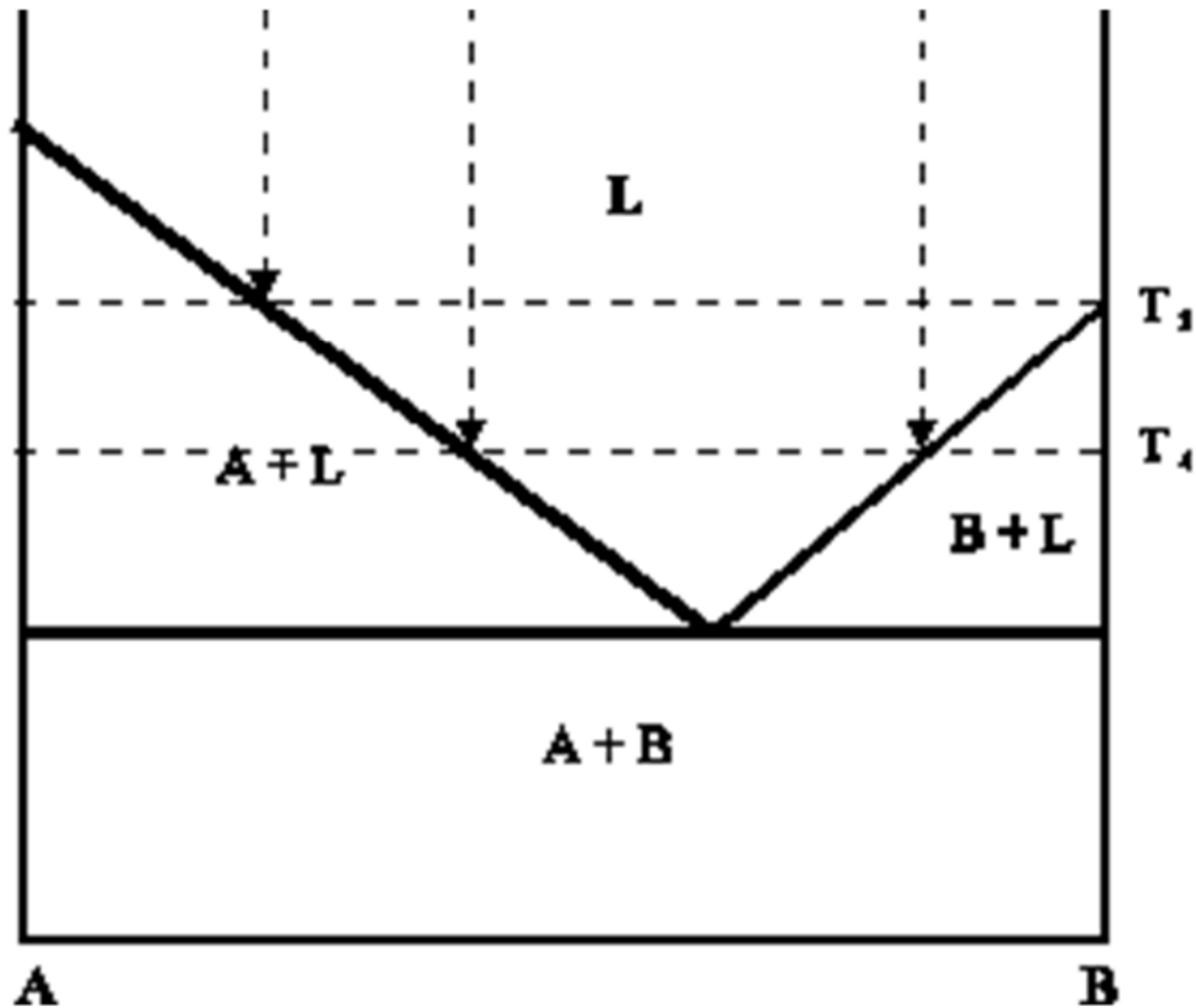
Phases: 1 phase solid

Composition: 80% Cr_2O_3 , 20% Al_2O_3

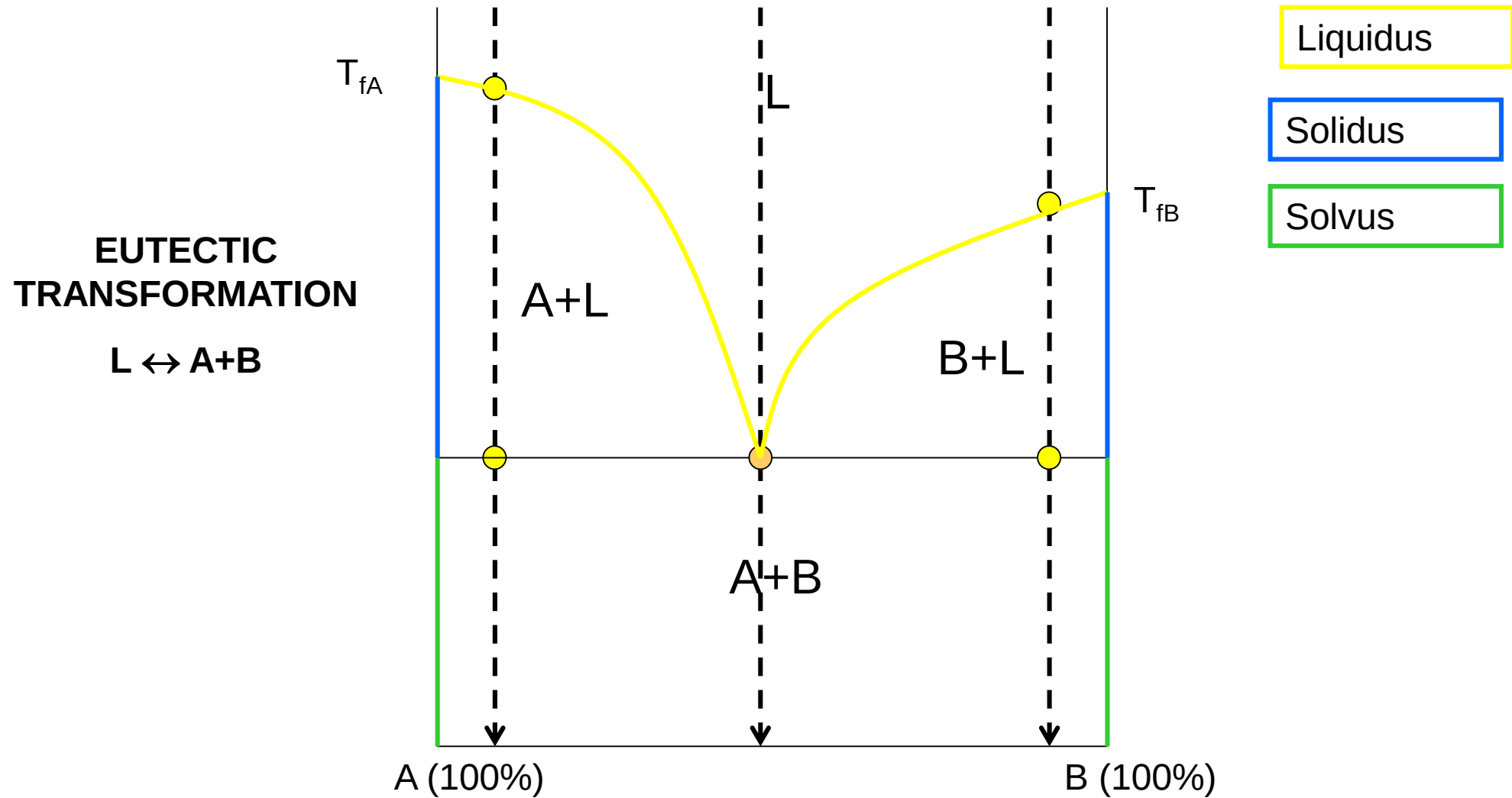
Relative amount: 100% solid phase

Exercise 5: Draw a phase diagram (for generic components A and B), with **complete solubility in the liquid state** and **no solubility in the solid state**. Also indicate the phase/phases present(s) in each region of the phase diagram

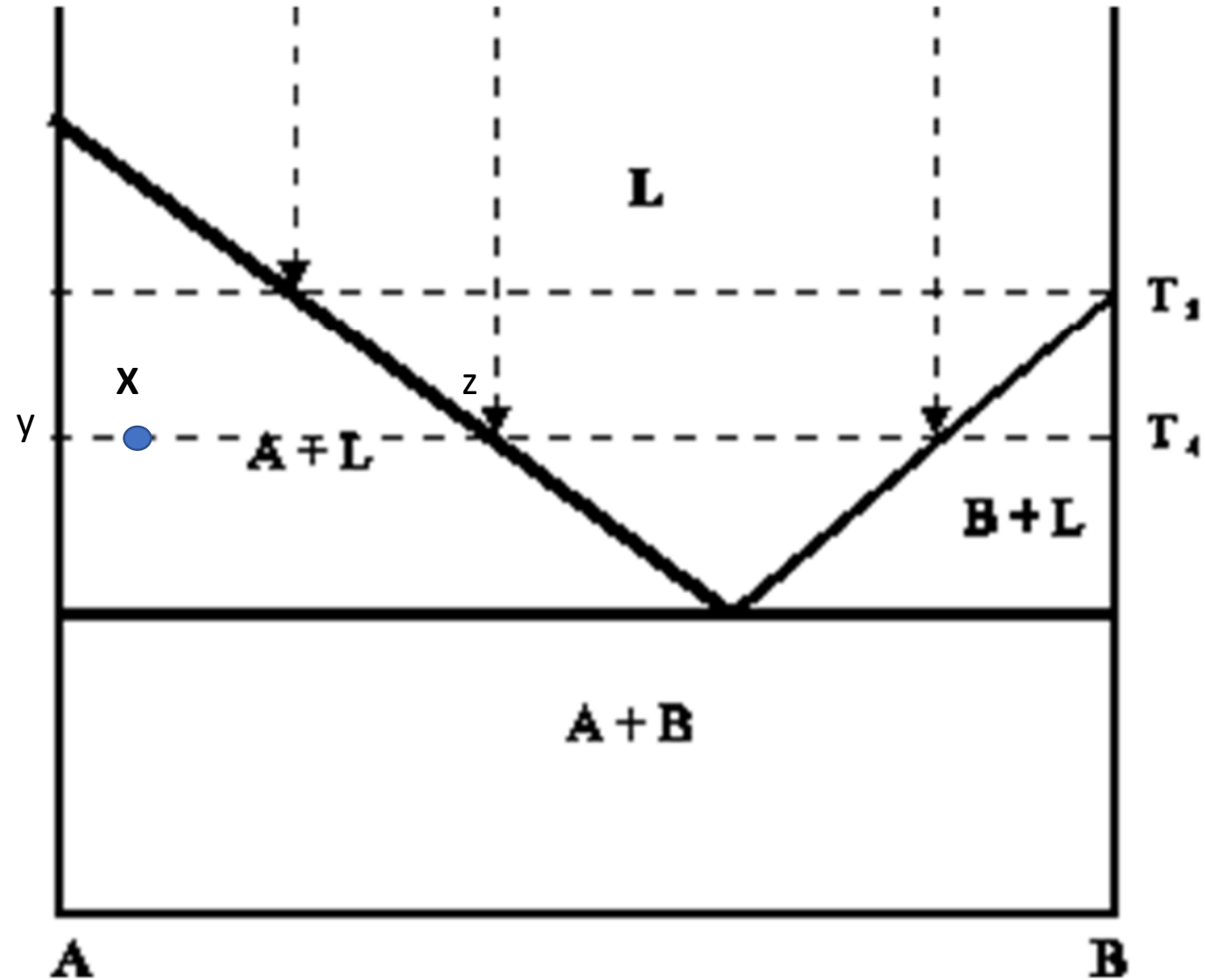
Complete solubility in the liquid state, **no solubility in the solid state**



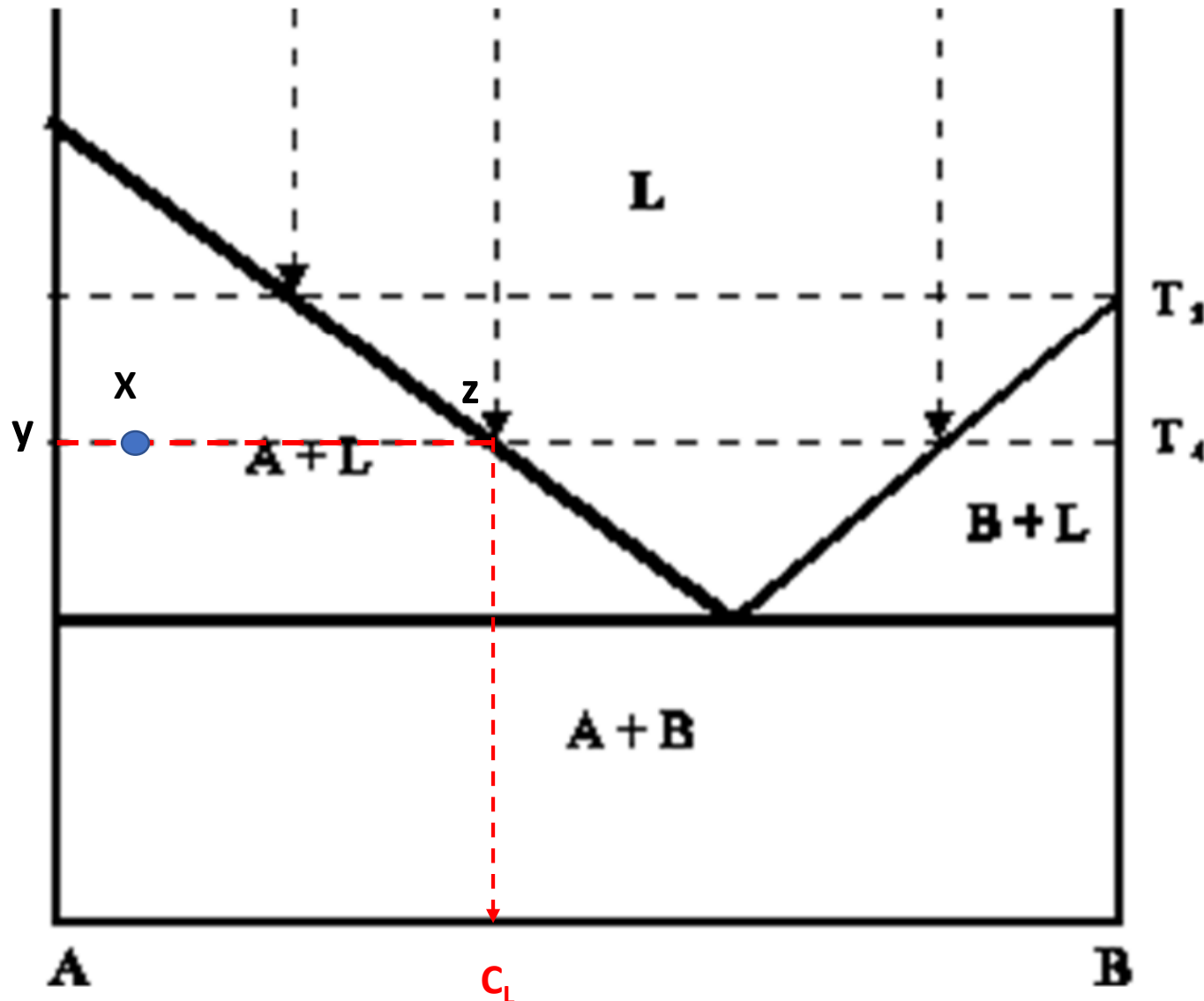
COMPLETE SOLUBILITY IN THE LIQUID STATE AND NO SOLUBILITY IN THE SOLID STATE, EUTECTIC



Exercise 6: determine the phases present, their compositions and relative amounts in the point X.



Exercise 6: determine the phases present, their compositions and relative amounts in the point X.



Point X

Phases: solid A+ Liquid L

Compositions: 2 phases region → horizontal rule

C_A : 100% A

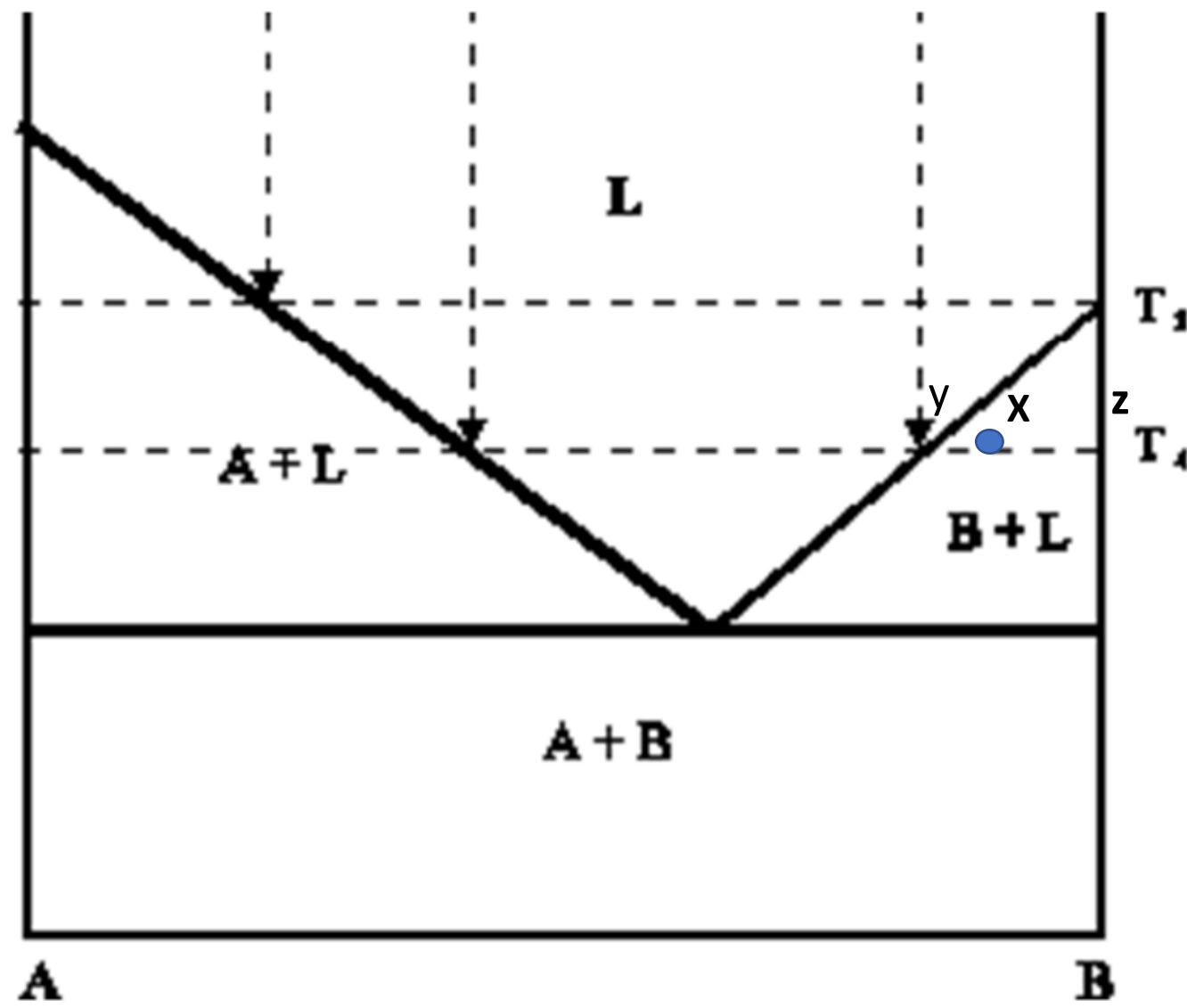
C_L : 45%B, 55%A

Relative amounts

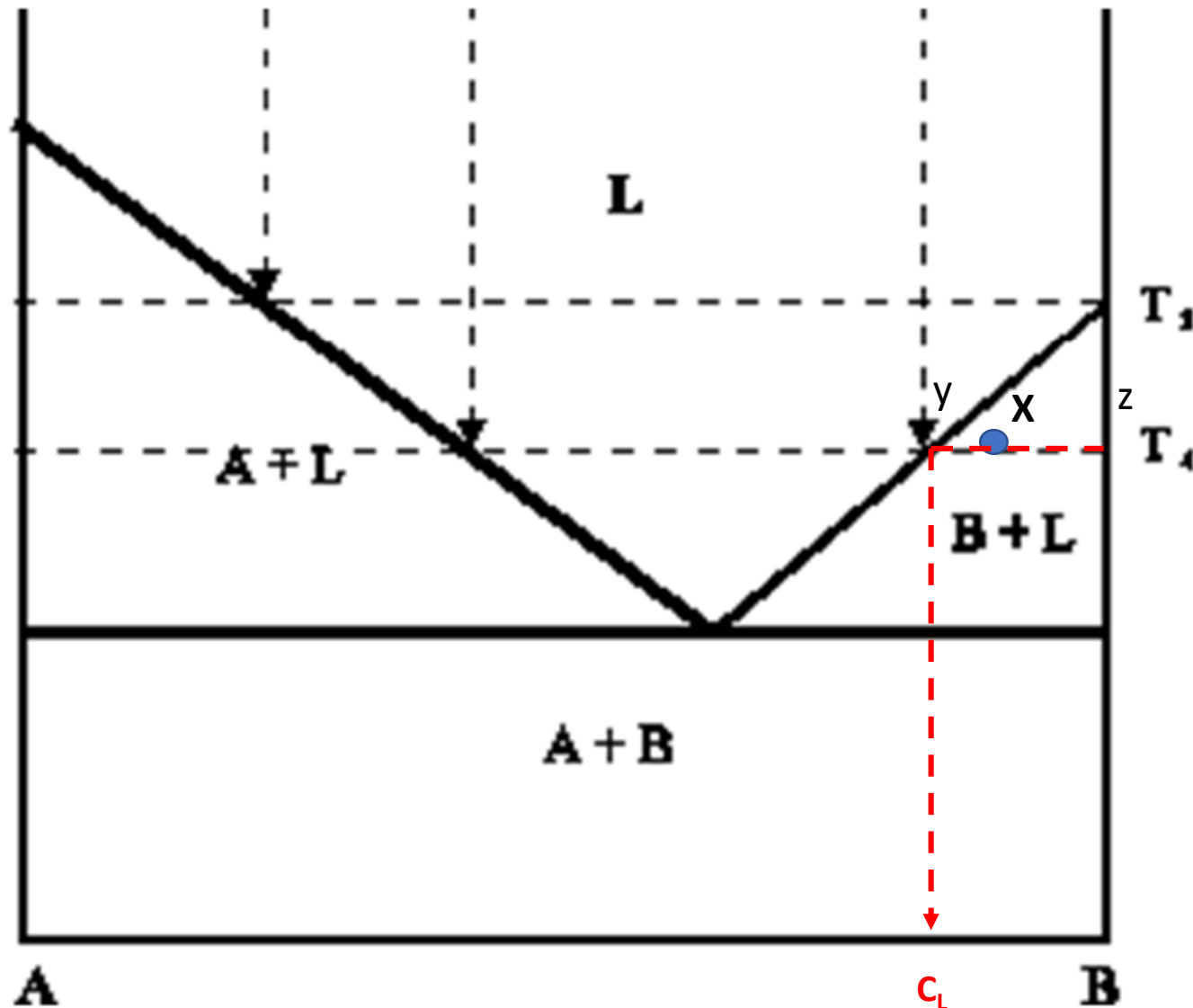
$\%A = (Xz)/(yz) * 100 = (45-10)/(45-0) * 100 = 78\%$

$\%L = (yX)/(yz) = (10-0)/(45-0) * 100 = 22\%$

Exercise 7: determine the phases present, their compositions and relative amounts in the point X.



Exercise 7: determine the phases present, their compositions and relative amounts in the point X.



Point X

Phases: solid B+ Liquid L

Compositions: 2 phases region → horizontal rule

C_B : 100% B

C_L : 80%B, 20%A

Relative amounts

$\%B = (Xy)/(yz) * 100 = (85-80)/(100-80) * 100 = 25\%$

$\%L = (Xz)/(yz) = (100-85)/(100-80) * 100 = 75\%$